Dual Simplex :-

Duality :-

To every LPP there corresponds another LPP called its dual.

Dual Simplex Method :-

1) In this method the starts from/condition is optimal but infeasible.
2) In Simplex method starts from/condition is feasible but not optimal.

Relationship b/w Simplex method and Dual Simplex method :-

Simplex	Dual
1) First we find entering Variable.	1) First we find Leaving Variable.
2) Take ratio of solution column with entering column.	2) Take ratio of Z row with Leaving row.
3) While taking ratio only positive entries are considered.	3) While taking ratio only negative entries

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	are considered.
4) Then we find leaving variable with least ratio.	4) Here we find entering variable with least ratio.
5) Then the table is revised and process continues until optimality reached.	5) Then the table is revised and the process continues until feasibility reached.

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Properties of the dual simplex method:

The dual simplex method starts with a dual feasible but primal infeasible solution and moves towards a primal and dual feasible optimal solution by maintaining dual feasibility while improving primal feasibility.

Its ability to handle problems where a dual feasible solution is easier to find, its effectiveness in re-optimizing after constraints are added, its improved numerical stability due to flexibility in pivot selection, and its ability to handle degeneracy well. The method selects the leaving variable first (based on the most negative primal infeasible variable) and then the entering variable (based on a ratio test with negative co-efficients.

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Advantage of Dual Simplex Algorithm:

1) Suitable for problems where it is easier to find a dual feasible solution than a primal feasible one.
2) Efficient in re-optimizing problems after small changes to constraints.
3) If you have a feasible dual solution but an infeasible primal, the dual simplex method can restore ~~feasibility~~ feasibility without restarting the entire optimization.
4) This often happens after adding new constraints.

Disadvantage of the dual simplex:

1) Finding a dual-feasible starting point can be difficult compared to finding a primal-feasible one which is what the regular simplex method uses.

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2) The pivot rules and interpretation are more complex than in the Primal simplex method.

3) The algorithm can be more sensitive to rounding errors due to handling infeasible Primal solutions.

4) Choosing the correct leaving and entering variables (pivots) requires careful consideration to maintain dual feasibility.

Characteristics of the Dual Simplex:

1) At every iteration, the dual constraints remain satisfied.

2) That means all the reduced costs for non-basic variable) satisfy the dual feasibility conditions.

3) The algorithm starts with an infeasible Primal solution and moves toward feasibility.

4) The goal is to make the basic variables non-negative while maintaining dual feasibility.

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Condition of dual simplex: To apply the dual simplex method, we need to first convert our LP problem, such The objective is expressed as a maximization. All constraints are in the form of "$\leq$" except for the positivity constraints. After that, we standardize the converted LP into standard form.

Dual simplex communication: There is a technical distinction between full-duplex communication, which uses a single physical communication channel for both directions simultaneously and dual simplex communication which uses two distinct channels, one for each direction.

Dual and primal: The term "Duality" implies that every linear programming problem, whether of maximization or minimization, is associated with another linear programming problem based on the same data. The original problem in a LPP is called the primal problem, whereas the other is called its dual problem.

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